ADSP 31014 Statistical Models for Data Science
Course Project Part 2

Business Problem

The City of Chicago HR Department wants to understand the salary structure of salaried employees and identify key factors that influence annual compensation levels across different departments and positions.

Linear Regression Model

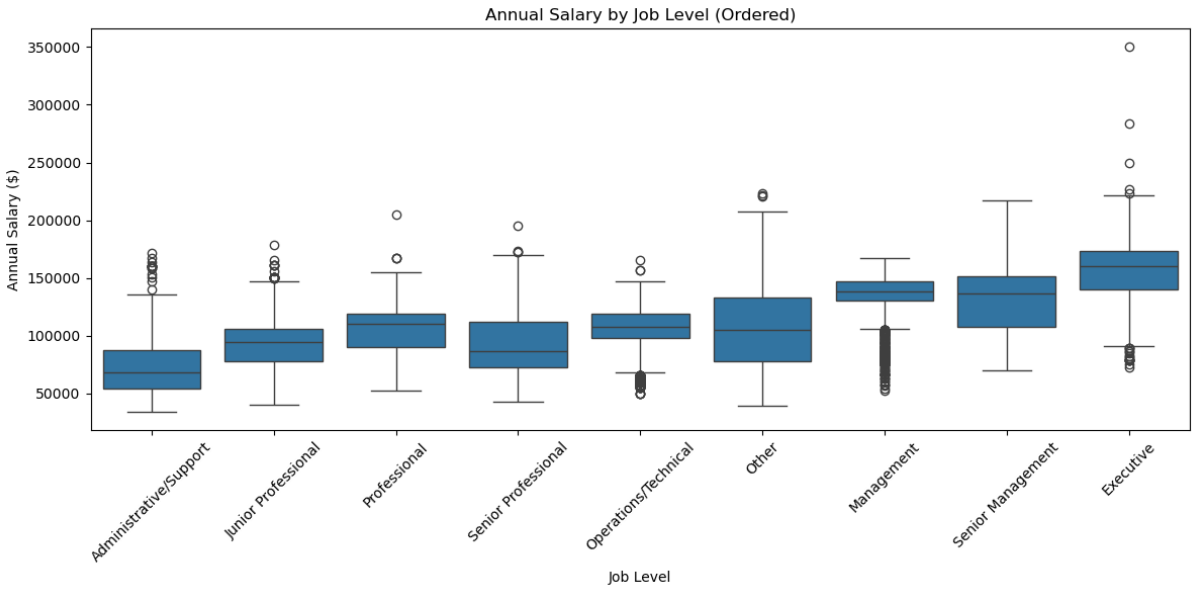
We build a linear regression model using 25,009 full-time salaried employees from The City of Chicago. We use log(annual_salary) as the response variable with job hierarchical level and department as explanatory variables. The model formula is:

$$\log(annual\ salary) \sim department + job\ level + department * job\ level$$

This model has 351 parameters with $R^2 = 0.52$. See Appendix for model summary.

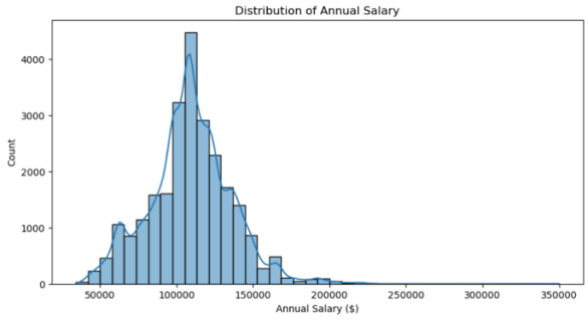
Key Considerations in Modeling

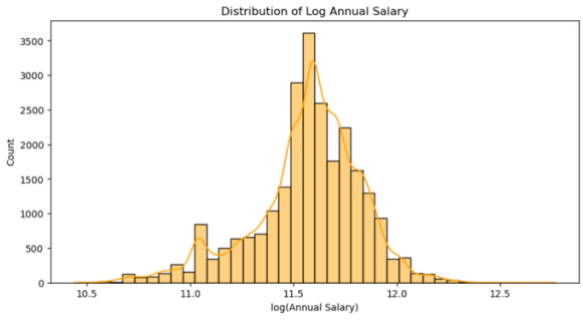
- Feature Engineering (Job Level):
  With more than 1,000 unique job titles, a job-level hierarchy was engineered to represent responsibility and seniority. Job_titles were grouped into ten categories such as Executive, Senior Management, Management, Professional, and Administrative/Support. The engineered job_level variable exhibits clear salary separation across levels, supporting its validity.



- Transformation of the Response Variable:
  Salaries show a right-skewed distribution because high-level executive roles are few in number but have significantly higher pay, creating a long tail on the upper

end. Applying a log transformation yields a more symmetric distribution and reduces heteroscedasticity. This improves the interpretability and statistical validity of the linear model.

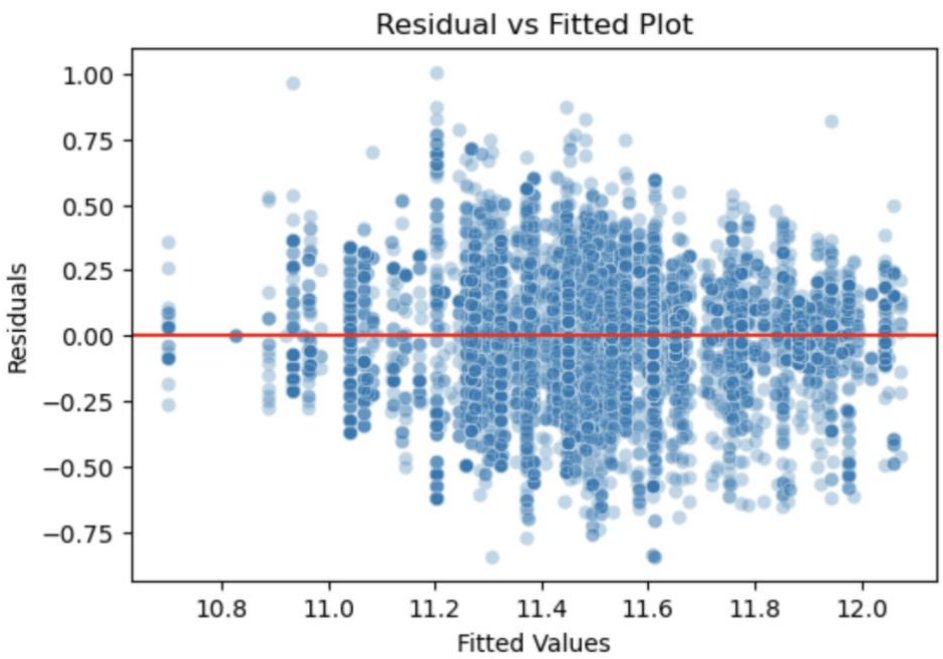




- Model Selection: Baseline vs Interaction
  - Model 1 (Baseline): $R^2 = 0.4424$, Adj. $R^2 = 0.4414$, AIC = -10,210, BIC = -9,828
  - Model 2 (Interaction): $R^2 = 0.5188$, Adj. $R^2 = 0.5135$, AIC = -13,446, BIC = -11,251
  - F-test: $F = 17.59$, $p < 0.001$ (large model (model 2) is better)

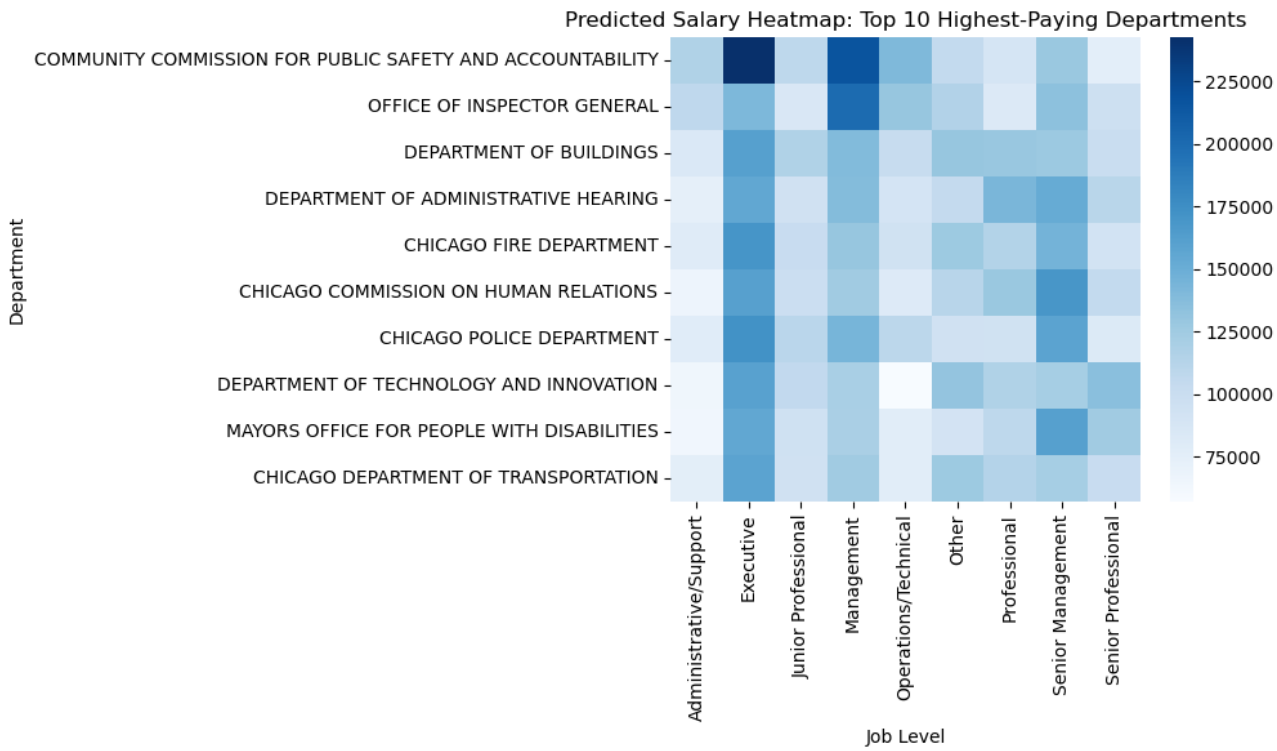
  The interaction model (model 2) substantially improves adjusted R^2 and lowers AIC/BIC, indicating better goodness-of-fit without excessive overfitting.

- Model Diagnostics:
  Plot of Residual plots indicate no major violations of linearity. Heteroscedasticity is modest and acceptable after log transformation.



Other Insights

- A heatmap of predicted salary across the five highest-paying departments and all job levels highlights clear compensation structures. It also reflects different market rates and organizational complexity.



- Key findings include:
  - Executives and Management receive the highest predicted salaries across departments, while Administrative/Support receive the lowest.
  - Salary gaps between departments are often larger than salary gaps across job levels. Even within the same job level, predicted pay varies substantially by department.
  - Some departments (e.g., Technology & Innovation) show competitive salaries even at mid-level roles (e.g., Senior Professional). This suggests that their compensation structures are not solely concentrated at the executive tier.

Model Shortcomings

- Missing key salary determinants:
  The dataset does not include important drivers such as years of experience, education level, and performance ratings, which introduces potential omitted variable bias.
- Sparse combinations in interaction terms:
  The full Department × Job Level interaction creates many small-sample groups, causing unstable or insignificant coefficients and limiting the reliability of some interaction effects.
- Moderate explanatory power:
  Even with interactions, the model explains only about 52% of salary variation, suggesting substantial unobserved heterogeneity (specialized skills, tenure, union steps, individual negotiation).

Appendix 1

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                                 OLS Regression Results
==============================================================================
Dep. Variable:            log_salary   R-squared:                       0.519
Model:                           OLS   Adj. R-squared:                  0.514
Method:                Least Squares   F-statistic:                     99.14
Date:               Mon, 24 Nov 2025   Prob (F-statistic):               0.00
Time:                       03:51:42   Log-Likelihood:                 6992.9
No. Observations:              25009   AIC:                        -1.345e+04
Df Residuals:                  24739   BIC:                        -1.125e+04
Df Model:                        269
Covariance Type:           nonrobust
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==========================================================================

coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
--------------------------------------------------------------------------
Intercept
10.6997      0.035    302.251      0.000      10.630      10.769
C(department)[T.BOARD OF ETHICS]
0.3362      0.074      4.548      0.000       0.191       0.481
C(department)[T.CHICAGO ANIMAL CARE AND CONTROL]
0.4179      0.078      5.356      0.000       0.265       0.571
C(department)[T.CHICAGO COMMISSION ON HUMAN RELATIONS]
0.4116      0.051      8.025      0.000       0.311       0.512
C(department)[T.CHICAGO DEPARTMENT OF AVIATION]
0.2674      0.046      5.836      0.000       0.178       0.357
C(department)[T.CHICAGO DEPARTMENT OF PUBLIC HEALTH]
0.3730      0.054      6.873      0.000       0.267       0.479
C(department)[T.CHICAGO DEPARTMENT OF TRANSPORTATION]
0.5442      0.053     10.301      0.000       0.441       0.648
C(department)[T.CHICAGO FIRE DEPARTMENT]
0.5885      0.112      5.257      0.000       0.369       0.808
C(department)[T.CHICAGO POLICE BOARD]
0.2653      0.090      2.958      0.003       0.089       0.441
C(department)[T.CHICAGO POLICE DEPARTMENT]
0.5689      0.037     15.340      0.000       0.496       0.642
C(department)[T.CHICAGO PUBLIC LIBRARY]
0.3403      0.037      9.149      0.000       0.267       0.413
C(department)[T.CITY COUNCIL]
0.6072      0.041     14.682      0.000       0.526       0.688
C(department)[T.CITY TREASURER'S OFFICE]
0.3073      0.051      5.991      0.000       0.207       0.408
C(department)[T.CIVILIAN OFFICE OF POLICE ACCOUNTABILITY]
0.5080      0.135      3.769      0.000       0.244       0.772
C(department)[T.COMMUNITY COMMISSION FOR PUBLIC SAFETY AND ACCOUNTABILITY]
0.9616      0.187      5.134      0.000       0.594       1.329
C(department)[T.DEPARTMENT OF ADMINISTRATIVE HEARING]
0.5157      0.059      8.706      0.000       0.400       0.632
C(department)[T.DEPARTMENT OF BUILDINGS]
0.6443      0.062     10.376      0.000       0.523       0.766
C(department)[T.DEPARTMENT OF BUSINESS AFFAIRS AND CONSUMER PROTECTION]
0.4430      0.135      3.287      0.001       0.179       0.707
C(department)[T.DEPARTMENT OF CULTURAL AFFAIRS AND SPECIAL EVENTS]
0.2319      0.040      5.775      0.000       0.153       0.311
C(department)[T.DEPARTMENT OF ENVIRONMENT]
0.1692      0.061      2.754      0.006       0.049       0.290
C(department)[T.DEPARTMENT OF FAMILY AND SUPPORT SERVICES]
0.4452      0.057      7.817      0.000       0.334       0.557
C(department)[T.DEPARTMENT OF FINANCE]
0.2334      0.040      5.879      0.000       0.156       0.311
C(department)[T.DEPARTMENT OF FLEET AND FACILITY MANAGEMENT]
0.5044      0.066      7.665      0.000       0.375       0.633
C(department)[T.DEPARTMENT OF HOUSING]
0.3592      0.187      1.918      0.055      -0.008       0.726
C(department)[T.DEPARTMENT OF HUMAN RESOURCES]
0.6788      0.083      8.176      0.000       0.516       0.842
C(department)[T.DEPARTMENT OF LAW]
0.4440      0.074      5.997      0.000       0.299       0.589
C(department)[T.DEPARTMENT OF PLANNING AND DEVELOPMENT]
0.3701      0.112      3.306      0.001       0.151       0.589
C(department)[T.DEPARTMENT OF PROCUREMENT SERVICES]
0.2851      0.090      3.183      0.001       0.110       0.461
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C(department)[T.DEPARTMENT OF STREETS AND SANITATION]
0.4207 0.046 9.183 0.000 0.331 0.511
C(department)[T.DEPARTMENT OF TECHNOLOGY AND INNOVATION]
0.3744 0.039 9.692 0.000 0.299 0.450
C(department)[T.DEPARTMENT OF WATER MANAGEMENT]
0.3817 0.051 7.476 0.000 0.282 0.482
C(department)[T.LICENSE APPEAL COMMISSION]
0.1656 0.095 1.735 0.083 -0.021 0.353
C(department)[T.MAYORS OFFICE FOR PEOPLE WITH DISABILITIES]
0.3670 0.055 6.639 0.000 0.259 0.475
C(department)[T.OFFICE OF BUDGET & MANAGEMENT]
0.2920 0.046 6.393 0.000 0.202 0.381
C(department)[T.OFFICE OF CITY CLERK]
0.6030 0.062 9.710 0.000 0.481 0.725
C(department)[T.OFFICE OF EMERGENCY MANAGEMENT AND COMMUNICATIONS]
0.3674 0.041 9.028 0.000 0.288 0.447
C(department)[T.OFFICE OF INSPECTOR GENERAL]
0.8828 0.090 9.858 0.000 0.707 1.058
C(department)[T.OFFICE OF PUBLIC SAFETY ADMINISTRATION]
0.5451 0.083 6.566 0.000 0.382 0.708
C(department)[T.OFFICE OF THE MAYOR]
0.3472 0.038 9.078 0.000 0.272 0.422
C(job_level)[T.Executive]
0.7380 0.021 35.319 0.000 0.697 0.779
C(job_level)[T.Junior Professional]
0.1893 0.058 3.262 0.001 0.076 0.303
C(job_level)[T.Management]
0.6263 0.099 6.355 0.000 0.433 0.819
C(job_level)[T.Operations/Technical]
0.1931 0.027 7.263 0.000 0.141 0.245
C(job_level)[T.Other]
0.6274 0.062 10.103 0.000 0.506 0.749
C(job_level)[T.Professional]
0.6528 0.187 3.485 0.000 0.286 1.020
C(job_level)[T.Senior Management]
0.9289 0.071 13.120 0.000 0.790 1.068
C(job_level)[T.Senior Professional]
0.2631 0.048 5.474 0.000 0.169 0.357
C(department)[T.BOARD OF ETHICS]:C(job_level)[T.Executive]
4.764e-15 1.94e-15 2.452 0.014 9.56e-16 8.57e-15
C(department)[T.CHICAGO ANIMAL CARE AND CONTROL]:C(job_level)[T.Executive]
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0.1392 0.113 1.232 0.218 -0.082 0.361
C(department)[T.CHICAGO DEPARTMENT OF AVIATION]:C(job_level)[T.Executive]
0.2396 0.040 5.995 0.000 0.161 0.318
C(department)[T.CHICAGO DEPARTMENT OF PUBLIC HEALTH]:C(job_level)[T.Executive]
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-0.0057 0.054 -0.105 0.916 -0.112 0.101
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0.0186 0.105 0.176 0.860 -0.188 0.225
C(department)[T.CHICAGO POLICE BOARD]:C(job_level)[T.Executive]
-1.225e-14 2.46e-15 -4.983 0.000 -1.71e-14 -7.43e-15
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0.0532 0.039 1.357 0.175 -0.024 0.130
C(department)[T.CHICAGO PUBLIC LIBRARY]:C(job_level)[T.Executive]
0.0608 0.048 1.264 0.206 -0.033 0.155
C(department)[T.CITY COUNCIL]:C(job_level)[T.Executive]
-0.2969 0.180 -1.649 0.099 -0.650 0.056
C(department)[T.CITY TREASURER'S OFFICE]:C(job_level)[T.Executive]
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C(department)[T.CIVILIAN OFFICE OF POLICE ACCOUNTABILITY]:C(job_level)[T.Executive]
-0.0083 0.164 -0.051 0.959 -0.329 0.312
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-1.344e-15 1.1e-15 -1.219 0.223 -3.51e-15 8.18e-16
C(department)[T.DEPARTMENT OF ADMINISTRATIVE HEARING]:C(job_level)[T.Executive]
-1.893e-15 8.32e-16 -2.276 0.023 -3.52e-15 -2.62e-16
C(department)[T.DEPARTMENT OF BUILDINGS]:C(job_level)[T.Executive]
-0.0889 0.070 -1.278 0.201 -0.225 0.047
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0.0810 0.134 0.604 0.546 -0.182 0.344
C(department)[T.DEPARTMENT OF CULTURAL AFFAIRS AND SPECIAL EVENTS]:C(job_level)[T.Executive]
0.1728 0.070 2.485 0.013 0.037 0.309
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0.3874 0.121 3.192 0.001 0.149 0.625

C(department)[T.DEPARTMENT OF FAMILY AND SUPPORT SERVICES]:C(job_level)[T.Executive]
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C(department)[T.DEPARTMENT OF FINANCE]:C(job_level)[T.Executive]
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0.1332 0.186 0.718 0.473 -0.231 0.497
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0.1612 0.076 2.114 0.035 0.012 0.311
C(department)[T.DEPARTMENT OF PLANNING AND DEVELOPMENT]:C(job_level)[T.Executive]
0.1317 0.111 1.191 0.233 -0.085 0.348
C(department)[T.DEPARTMENT OF PROCUREMENT SERVICES]:C(job_level)[T.Executive]
0.0532 0.131 0.405 0.685 -0.204 0.311
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-0.1005 0.041 -2.445 0.014 -0.181 -0.020
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0.1723 0.088 1.954 0.051 -0.000 0.345
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3.307e-15 1.27e-15 2.614 0.009 8.27e-16 5.79e-15
C(department)[T.MAYORS OFFICE FOR PEOPLE WITH DISABILITIES]:C(job_level)[T.Executive]
0.1478 0.119 1.246 0.213 -0.085 0.380
C(department)[T.OFFICE OF BUDGET & MANAGEMENT]:C(job_level)[T.Executive]
-0.0839 0.159 -0.527 0.598 -0.396 0.228
C(department)[T.OFFICE OF CITY CLERK]:C(job_level)[T.Executive]
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C(department)[T.OFFICE OF EMERGENCY MANAGEMENT AND COMMUNICATIONS]:C(job_level)[T.Executive]
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-0.3072 0.096 -3.202 0.001 -0.495 -0.119
C(department)[T.OFFICE OF THE MAYOR]:C(job_level)[T.Executive]
0.2874 0.064 4.522 0.000 0.163 0.412
C(department)[T.BOARD OF ETHICS]:C(job_level)[T.Junior Professional]
0.3687 0.153 2.408 0.016 0.069 0.669
C(department)[T.CHICAGO ANIMAL CARE AND CONTROL]:C(job_level)[T.Junior Professional]
-0.2390 0.105 -2.276 0.023 -0.445 -0.033
C(department)[T.CHICAGO COMMISSION ON HUMAN RELATIONS]:C(job_level)[T.Junior Professional]
0.1976 0.078 2.526 0.012 0.044 0.351
C(department)[T.CHICAGO DEPARTMENT OF AVIATION]:C(job_level)[T.Junior Professional]
0.3655 0.074 4.921 0.000 0.220 0.511
C(department)[T.CHICAGO DEPARTMENT OF PUBLIC HEALTH]:C(job_level)[T.Junior Professional]
0.2249 0.072 3.118 0.002 0.084 0.366
C(department)[T.CHICAGO DEPARTMENT OF TRANSPORTATION]:C(job_level)[T.Junior Professional]
0.0239 0.074 0.324 0.746 -0.121 0.168
C(department)[T.CHICAGO FIRE DEPARTMENT]:C(job_level)[T.Junior Professional]
0.0543 0.122 0.447 0.655 -0.184 0.292
C(department)[T.CHICAGO POLICE BOARD]:C(job_level)[T.Junior Professional]
0.0917 0.142 0.648 0.517 -0.186 0.369
C(department)[T.CHICAGO POLICE DEPARTMENT]:C(job_level)[T.Junior Professional]
0.1506 0.064 2.347 0.019 0.025 0.276
C(department)[T.CHICAGO PUBLIC LIBRARY]:C(job_level)[T.Junior Professional]
0.1502 0.060 2.501 0.012 0.033 0.268
C(department)[T.CITY COUNCIL]:C(job_level)[T.Junior Professional]
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C(department)[T.CITY TREASURER'S OFFICE]:C(job_level)[T.Junior Professional]
0.1534 0.124 1.240 0.215 -0.089 0.396
C(department)[T.CIVILIAN OFFICE OF POLICE ACCOUNTABILITY]:C(job_level)[T.Junior Professional]
-0.0672 0.161 -0.417 0.676 -0.383 0.248
C(department)[T.COMMUNITY COMMISSION FOR PUBLIC SAFETY AND ACCOUNTABILITY]:C(job_level)[T.Junior Professional] -0.2567 0.267 -0.963 0.335 -0.779 0.266
C(department)[T.DEPARTMENT OF ADMINISTRATIVE HEARING]:C(job_level)[T.Junior Professional]
0.0499 0.199 0.251 0.802 -0.339 0.439
C(department)[T.DEPARTMENT OF BUILDINGS]:C(job_level)[T.Junior Professional]
0.1290 0.120 1.074 0.283 -0.106 0.364
C(department)[T.DEPARTMENT OF BUSINESS AFFAIRS AND CONSUMER PROTECTION]:C(job_level)[T.Junior Professional] 0.3046 0.178 1.714 0.086 -0.044 0.653
C(department)[T.DEPARTMENT OF CULTURAL AFFAIRS AND SPECIAL EVENTS]:C(job_level)[T.Junior Professional]
0.2118 0.066 3.226 0.001 0.083 0.340
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-2.795e-15 1.28e-15 -2.176 0.030 -5.31e-15 -2.77e-16

C(department)[T.DEPARTMENT OF FAMILY AND SUPPORT SERVICES]:C(job_level)[T.Junior Professional]
0.0868 0.078 1.117 0.264 -0.066 0.239
C(department)[T.DEPARTMENT OF FINANCE]:C(job_level)[T.Junior Professional]
0.3353 0.067 5.011 0.000 0.204 0.466
C(department)[T.DEPARTMENT OF FLEET AND FACILITY MANAGEMENT]:C(job_level)[T.Junior Professional]
-0.0190 0.089 -0.213 0.831 -0.194 0.156
C(department)[T.DEPARTMENT OF HOUSING]:C(job_level)[T.Junior Professional]
0.1607 0.207 0.777 0.437 -0.245 0.566
C(department)[T.DEPARTMENT OF HUMAN RESOURCES]:C(job_level)[T.Junior Professional]
-0.3933 0.126 -3.132 0.002 -0.639 -0.147
C(department)[T.DEPARTMENT OF LAW]:C(job_level)[T.Junior Professional]
0.1930 0.088 2.187 0.029 0.020 0.366
C(department)[T.DEPARTMENT OF PLANNING AND DEVELOPMENT]:C(job_level)[T.Junior Professional]
0.1112 0.127 0.874 0.382 -0.138 0.361
C(department)[T.DEPARTMENT OF PROCUREMENT SERVICES]:C(job_level)[T.Junior Professional]
-3.347e-15 1.46e-15 -2.286 0.022 -6.22e-15 -4.77e-16
C(department)[T.DEPARTMENT OF STREETS AND SANITATION]:C(job_level)[T.Junior Professional]
0.1839 0.080 2.286 0.022 0.026 0.342
C(department)[T.DEPARTMENT OF TECHNOLOGY AND INNOVATION]:C(job_level)[T.Junior Professional]
0.3083 0.107 2.877 0.004 0.098 0.518
C(department)[T.DEPARTMENT OF WATER MANAGEMENT]:C(job_level)[T.Junior Professional]
0.2172 0.072 3.022 0.003 0.076 0.358
C(department)[T.LICENSE APPEAL COMMISSION]:C(job_level)[T.Junior Professional]
-1.255e-15 1.17e-15 -1.076 0.282 -3.54e-15 1.03e-15
C(department)[T.MAYORS OFFICE FOR PEOPLE WITH DISABILITIES]:C(job_level)[T.Junior Professional]
0.2083 0.083 2.500 0.012 0.045 0.372
C(department)[T.OFFICE OF BUDGET & MANAGEMENT]:C(job_level)[T.Junior Professional]
0.2964 0.108 2.750 0.006 0.085 0.508
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-0.0733 0.113 -0.649 0.516 -0.295 0.148
C(department)[T.OFFICE OF EMERGENCY MANAGEMENT AND COMMUNICATIONS]:C(job_level)[T.Junior Professional]
0.0679 0.062 1.095 0.273 -0.054 0.189
C(department)[T.OFFICE OF INSPECTOR GENERAL]:C(job_level)[T.Junior Professional]
-0.4171 0.109 -3.835 0.000 -0.630 -0.204
C(department)[T.OFFICE OF PUBLIC SAFETY ADMINISTRATION]:C(job_level)[T.Junior Professional]
0.0085 0.102 0.083 0.934 -0.192 0.209
C(department)[T.OFFICE OF THE MAYOR]:C(job_level)[T.Junior Professional]
-0.0287 0.095 -0.303 0.762 -0.214 0.157
C(department)[T.BOARD OF ETHICS]:C(job_level)[T.Management]
2.695e-16 3.95e-16 0.683 0.495 -5.04e-16 1.04e-15
C(department)[T.CHICAGO ANIMAL CARE AND CONTROL]:C(job_level)[T.Management]
-0.1127 0.142 -0.793 0.428 -0.391 0.166
C(department)[T.CHICAGO COMMISSION ON HUMAN RELATIONS]:C(job_level)[T.Management]
1.275e-15 5.48e-16 2.329 0.020 2.02e-16 2.35e-15
C(department)[T.CHICAGO DEPARTMENT OF AVIATION]:C(job_level)[T.Management]
-0.1430 0.104 -1.376 0.169 -0.347 0.061
C(department)[T.CHICAGO DEPARTMENT OF PUBLIC HEALTH]:C(job_level)[T.Management]
-0.2988 0.151 -1.984 0.047 -0.594 -0.004
C(department)[T.CHICAGO DEPARTMENT OF TRANSPORTATION]:C(job_level)[T.Management]
-0.1358 0.124 -1.092 0.275 -0.380 0.108
C(department)[T.CHICAGO FIRE DEPARTMENT]:C(job_level)[T.Management]
-0.1386 0.145 -0.956 0.339 -0.423 0.146
C(department)[T.CHICAGO POLICE BOARD]:C(job_level)[T.Management]
1.439e-16 2.79e-16 0.516 0.606 -4.02e-16 6.9e-16
C(department)[T.CHICAGO POLICE DEPARTMENT]:C(job_level)[T.Management]
-0.0126 0.099 -0.126 0.899 -0.207 0.182
C(department)[T.CHICAGO PUBLIC LIBRARY]:C(job_level)[T.Management]
-0.1722 0.129 -1.336 0.182 -0.425 0.080
C(department)[T.CITY COUNCIL]:C(job_level)[T.Management]
-0.6264 0.130 -4.814 0.000 -0.882 -0.371
C(department)[T.CITY TREASURER'S OFFICE]:C(job_level)[T.Management]
2.017e-16 2.6e-16 0.777 0.437 -3.07e-16 7.11e-16
C(department)[T.CIVILIAN OFFICE OF POLICE ACCOUNTABILITY]:C(job_level)[T.Management]
-1.36e-15 5.13e-16 -2.651 0.008 -2.37e-15 -3.55e-16
C(department)[T.COMMUNITY COMMISSION FOR PUBLIC SAFETY AND ACCOUNTABILITY]:C(job_level)[T.Management]
-6.964e-16 3.65e-16 -1.908 0.056 -1.41e-15 1.91e-17
C(department)[T.DEPARTMENT OF ADMINISTRATIVE HEARING]:C(job_level)[T.Management]
-1.545e-16 3.89e-16 -0.397 0.691 -9.17e-16 6.08e-16
C(department)[T.DEPARTMENT OF BUILDINGS]:C(job_level)[T.Management]
-0.1281 0.138 -0.927 0.354 -0.399 0.143
C(department)[T.DEPARTMENT OF BUSINESS AFFAIRS AND CONSUMER PROTECTION]:C(job_level)[T.Management]
-0.2214 0.173 -1.278 0.201 -0.561 0.118
C(department)[T.DEPARTMENT OF CULTURAL AFFAIRS AND SPECIAL EVENTS]:C(job_level)[T.Management]
-0.0806 0.125 -0.643 0.520 -0.326 0.165
C(department)[T.DEPARTMENT OF ENVIRONMENT]:C(job_level)[T.Management]
7.731e-16 3.95e-16 1.957 0.050 -1.2e-18 1.55e-15

C(department)[T.DEPARTMENT OF FAMILY AND SUPPORT SERVICES]:C(job_level)[T.Management]
-0.2822 0.118 -2.401 0.016 -0.513 -0.052
C(department)[T.DEPARTMENT OF FINANCE]:C(job_level)[T.Management]
-0.0549 0.104 -0.529 0.597 -0.258 0.149
C(department)[T.DEPARTMENT OF FLEET AND FACILITY MANAGEMENT]:C(job_level)[T.Management]
-0.6207 0.172 -3.602 0.000 -0.959 -0.283
C(department)[T.DEPARTMENT OF HOUSING]:C(job_level)[T.Management]
3.427e-17 2.55e-16 0.135 0.893 -4.65e-16 5.33e-16
C(department)[T.DEPARTMENT OF HUMAN RESOURCES]:C(job_level)[T.Management]
-0.7062 0.137 -5.159 0.000 -0.974 -0.438
C(department)[T.DEPARTMENT OF LAW]:C(job_level)[T.Management]
0.1299 0.121 1.078 0.281 -0.106 0.366
C(department)[T.DEPARTMENT OF PLANNING AND DEVELOPMENT]:C(job_level)[T.Management]
0.0367 0.234 0.157 0.875 -0.422 0.496
C(department)[T.DEPARTMENT OF PROCUREMENT SERVICES]:C(job_level)[T.Management]
-4.897e-16 4.63e-16 -1.057 0.291 -1.4e-15 4.18e-16
C(department)[T.DEPARTMENT OF STREETS AND SANITATION]:C(job_level)[T.Management]
-0.0900 0.109 -0.828 0.408 -0.303 0.123
C(department)[T.DEPARTMENT OF TECHNOLOGY AND INNOVATION]:C(job_level)[T.Management]
-2.535e-16 1.44e-16 -1.765 0.078 -5.35e-16 2.8e-17
C(department)[T.DEPARTMENT OF WATER MANAGEMENT]:C(job_level)[T.Management]
-0.0437 0.149 -0.292 0.770 -0.337 0.249
C(department)[T.LICENSE APPEAL COMMISSION]:C(job_level)[T.Management]
1.187e-15 3.71e-16 3.200 0.001 4.6e-16 1.91e-15
C(department)[T.MAYORS OFFICE FOR PEOPLE WITH DISABILITIES]:C(job_level)[T.Management]
-3.038e-16 3.14e-16 -0.969 0.333 -9.18e-16 3.11e-16
C(department)[T.OFFICE OF BUDGET & MANAGEMENT]:C(job_level)[T.Management]
-1.185e-15 3.56e-16 -3.327 0.001 -1.88e-15 -4.87e-16
C(department)[T.OFFICE OF CITY CLERK]:C(job_level)[T.Management]
-0.4296 0.144 -2.981 0.003 -0.712 -0.147
C(department)[T.OFFICE OF EMERGENCY MANAGEMENT AND COMMUNICATIONS]:C(job_level)[T.Management]
-0.2628 0.118 -2.231 0.026 -0.494 -0.032
C(department)[T.OFFICE OF INSPECTOR GENERAL]:C(job_level)[T.Management]
1.169e-15 3.12e-16 3.742 0.000 5.57e-16 1.78e-15
C(department)[T.OFFICE OF PUBLIC SAFETY ADMINISTRATION]:C(job_level)[T.Management]
1.566e-16 2.95e-16 0.531 0.595 -4.21e-16 7.34e-16
C(department)[T.OFFICE OF THE MAYOR]:C(job_level)[T.Management]
1.444e-17 2.44e-16 0.059 0.953 -4.64e-16 4.93e-16
C(department)[T.BOARD OF ETHICS]:C(job_level)[T.Operations/Technical]
-6.239e-18 1.87e-16 -0.033 0.973 -3.73e-16 3.6e-16
C(department)[T.CHICAGO ANIMAL CARE AND CONTROL]:C(job_level)[T.Operations/Technical]
0.1620 0.187 0.869 0.385 -0.204 0.528
C(department)[T.CHICAGO COMMISSION ON HUMAN RELATIONS]:C(job_level)[T.Operations/Technical]
5.102e-16 1.99e-16 2.566 0.010 1.2e-16 9e-16
C(department)[T.CHICAGO DEPARTMENT OF AVIATION]:C(job_level)[T.Operations/Technical]
0.1135 0.039 2.896 0.004 0.037 0.190
C(department)[T.CHICAGO DEPARTMENT OF PUBLIC HEALTH]:C(job_level)[T.Operations/Technical]
-0.4394 0.098 -4.472 0.000 -0.632 -0.247
C(department)[T.CHICAGO DEPARTMENT OF TRANSPORTATION]:C(job_level)[T.Operations/Technical]
-0.1783 0.051 -3.507 0.000 -0.278 -0.079
C(department)[T.CHICAGO FIRE DEPARTMENT]:C(job_level)[T.Operations/Technical]
-0.0313 0.103 -0.303 0.762 -0.234 0.171
C(department)[T.CHICAGO POLICE BOARD]:C(job_level)[T.Operations/Technical]
-3.014e-16 2.04e-16 -1.479 0.139 -7.01e-16 9.82e-17
C(department)[T.CHICAGO POLICE DEPARTMENT]:C(job_level)[T.Operations/Technical]
0.1460 0.029 5.108 0.000 0.090 0.202
C(department)[T.CHICAGO PUBLIC LIBRARY]:C(job_level)[T.Operations/Technical]
0.0374 0.091 0.412 0.681 -0.141 0.216
C(department)[T.CITY COUNCIL]:C(job_level)[T.Operations/Technical]
-5.229e-16 3.72e-16 -1.406 0.160 -1.25e-15 2.06e-16
C(department)[T.CITY TREASURER'S OFFICE]:C(job_level)[T.Operations/Technical]
-2.006e-16 2.48e-16 -0.809 0.419 -6.87e-16 2.85e-16
C(department)[T.CIVILIAN OFFICE OF POLICE ACCOUNTABILITY]:C(job_level)[T.Operations/Technical]
3.036e-16 2.51e-16 1.211 0.226 -1.88e-16 7.95e-16
C(department)[T.COMMUNITY COMMISSION FOR PUBLIC SAFETY AND
ACCOUNTABILITY]:C(job_level)[T.Operations/Technical] 6.786e-16 2.06e-16 3.297 0.001
2.75e-16 1.08e-15
C(department)[T.DEPARTMENT OF ADMINISTRATIVE HEARING]:C(job_level)[T.Operations/Technical]
6.683e-16 2.02e-16 3.301 0.001 2.72e-16 1.07e-15
C(department)[T.DEPARTMENT OF BUILDINGS]:C(job_level)[T.Operations/Technical]
5.02e-17 1.16e-16 0.434 0.664 -1.77e-16 2.77e-16
C(department)[T.DEPARTMENT OF BUSINESS AFFAIRS AND CONSUMER
PROTECTION]:C(job_level)[T.Operations/Technical] 2.402e-16 2.74e-16 0.877 0.381 -
2.97e-16 7.77e-16
C(department)[T.DEPARTMENT OF CULTURAL AFFAIRS AND SPECIAL
EVENTS]:C(job_level)[T.Operations/Technical] -1.46e-15 4.3e-16 -3.395 0.001 -
2.3e-15 -6.17e-16

C(department)[T.DEPARTMENT OF ENVIRONMENT]:C(job_level)[T.Operations/Technical]
1.661e-16 1.56e-16 1.067 0.286 -1.39e-16 4.71e-16
C(department)[T.DEPARTMENT OF FAMILY AND SUPPORT SERVICES]:C(job_level)[T.Operations/Technical]
0.1193 0.111 1.071 0.284 -0.099 0.338
C(department)[T.DEPARTMENT OF FINANCE]:C(job_level)[T.Operations/Technical]
0.0442 0.044 1.016 0.310 -0.041 0.130
C(department)[T.DEPARTMENT OF FLEET AND FACILITY MANAGEMENT]:C(job_level)[T.Operations/Technical]
0.1806 0.104 1.732 0.083 -0.024 0.385
C(department)[T.DEPARTMENT OF HOUSING]:C(job_level)[T.Operations/Technical]
1.077e-16 1.65e-16 0.653 0.514 -2.16e-16 4.31e-16
C(department)[T.DEPARTMENT OF HUMAN RESOURCES]:C(job_level)[T.Operations/Technical]
8.318e-17 1.52e-16 0.546 0.585 -2.15e-16 3.82e-16
C(department)[T.DEPARTMENT OF LAW]:C(job_level)[T.Operations/Technical]
-0.4154 0.185 -2.244 0.025 -0.778 -0.053
C(department)[T.DEPARTMENT OF PLANNING AND DEVELOPMENT]:C(job_level)[T.Operations/Technical]
-0.0274 0.160 -0.172 0.864 -0.341 0.286
C(department)[T.DEPARTMENT OF PROCUREMENT SERVICES]:C(job_level)[T.Operations/Technical]
-1.02e-15 3.05e-16 -3.344 0.001 -1.62e-15 -4.22e-16
C(department)[T.DEPARTMENT OF STREETS AND SANITATION]:C(job_level)[T.Operations/Technical]
-3.187e-16 3.18e-16 -1.001 0.317 -9.43e-16 3.05e-16
C(department)[T.DEPARTMENT OF TECHNOLOGY AND INNOVATION]:C(job_level)[T.Operations/Technical]
-0.3155 0.118 -2.682 0.007 -0.546 -0.085
C(department)[T.DEPARTMENT OF WATER MANAGEMENT]:C(job_level)[T.Operations/Technical]
0.0563 0.078 0.717 0.473 -0.098 0.210
C(department)[T.LICENSE APPEAL COMMISSION]:C(job_level)[T.Operations/Technical]
7.032e-16 3.25e-16 2.166 0.030 6.68e-17 1.34e-15
C(department)[T.MAYORS OFFICE FOR PEOPLE WITH DISABILITIES]:C(job_level)[T.Operations/Technical]
-9.205e-17 1.63e-16 -0.566 0.571 -4.11e-16 2.27e-16
C(department)[T.OFFICE OF BUDGET & MANAGEMENT]:C(job_level)[T.Operations/Technical]
-1.842e-16 1.95e-16 -0.946 0.344 -5.66e-16 1.97e-16
C(department)[T.OFFICE OF CITY CLERK]:C(job_level)[T.Operations/Technical]
-3.394e-16 1.61e-16 -2.103 0.035 -6.56e-16 -2.31e-17
C(department)[T.OFFICE OF EMERGENCY MANAGEMENT AND
COMMUNICATIONS]:C(job_level)[T.Operations/Technical] 0.5386 0.044 12.312 0.000
0.453 0.624
C(department)[T.OFFICE OF INSPECTOR GENERAL]:C(job_level)[T.Operations/Technical]
-7.649e-17 1.39e-16 -0.549 0.583 -3.5e-16 1.97e-16
C(department)[T.OFFICE OF PUBLIC SAFETY ADMINISTRATION]:C(job_level)[T.Operations/Technical]
0.2025 0.115 1.767 0.077 -0.022 0.427
C(department)[T.OFFICE OF THE MAYOR]:C(job_level)[T.Operations/Technical]
-5.697e-17 1.56e-16 -0.364 0.716 -3.64e-16 2.5e-16
C(department)[T.BOARD OF ETHICS]:C(job_level)[T.Other]
0.1851 0.111 1.662 0.097 -0.033 0.403
C(department)[T.CHICAGO ANIMAL CARE AND CONTROL]:C(job_level)[T.Other]
-0.4569 0.099 -4.628 0.000 -0.650 -0.263
C(department)[T.CHICAGO COMMISSION ON HUMAN RELATIONS]:C(job_level)[T.Other]
-0.1164 0.082 -1.424 0.154 -0.277 0.044
C(department)[T.CHICAGO DEPARTMENT OF AVIATION]:C(job_level)[T.Other]
-0.2084 0.071 -2.946 0.003 -0.347 -0.070
C(department)[T.CHICAGO DEPARTMENT OF PUBLIC HEALTH]:C(job_level)[T.Other]
-0.0897 0.076 -1.182 0.237 -0.238 0.059
C(department)[T.CHICAGO DEPARTMENT OF TRANSPORTATION]:C(job_level)[T.Other]
-0.1183 0.075 -1.574 0.116 -0.266 0.029
C(department)[T.CHICAGO FIRE DEPARTMENT]:C(job_level)[T.Other]
-0.1607 0.129 -1.249 0.212 -0.413 0.091
C(department)[T.CHICAGO POLICE BOARD]:C(job_level)[T.Other]
0.1735 0.142 1.221 0.222 -0.105 0.452
C(department)[T.CHICAGO POLICE DEPARTMENT]:C(job_level)[T.Other]
-0.4415 0.065 -6.832 0.000 -0.568 -0.315
C(department)[T.CHICAGO PUBLIC LIBRARY]:C(job_level)[T.Other]
-0.1349 0.074 -1.826 0.068 -0.280 0.010
C(department)[T.CITY COUNCIL]:C(job_level)[T.Other]
-0.5631 0.067 -8.432 0.000 -0.694 -0.432
C(department)[T.CITY TREASURER'S OFFICE]:C(job_level)[T.Other]
-0.0188 0.075 -0.250 0.803 -0.167 0.129
C(department)[T.CIVILIAN OFFICE OF POLICE ACCOUNTABILITY]:C(job_level)[T.Other]
-0.2215 0.150 -1.477 0.140 -0.515 0.072
C(department)[T.COMMUNITY COMMISSION FOR PUBLIC SAFETY AND ACCOUNTABILITY]:C(job_level)[T.Other]
-0.7267 0.201 -3.610 0.000 -1.121 -0.332
C(department)[T.DEPARTMENT OF ADMINISTRATIVE HEARING]:C(job_level)[T.Other]
-0.2841 0.102 -2.794 0.005 -0.483 -0.085
C(department)[T.DEPARTMENT OF BUILDINGS]:C(job_level)[T.Other]
-0.1965 0.082 -2.409 0.016 -0.356 -0.037
C(department)[T.DEPARTMENT OF BUSINESS AFFAIRS AND CONSUMER PROTECTION]:C(job_level)[T.Other]
-0.3414 0.147 -2.326 0.020 -0.629 -0.054
C(department)[T.DEPARTMENT OF CULTURAL AFFAIRS AND SPECIAL EVENTS]:C(job_level)[T.Other]
0.1603 0.067 2.379 0.017 0.028 0.292

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C(department)[T.DEPARTMENT OF ENVIRONMENT]:C(job_level)[T.Other]
0.2692      0.129      2.082      0.037       0.016       0.523
C(department)[T.DEPARTMENT OF FAMILY AND SUPPORT SERVICES]:C(job_level)[T.Other]
-0.2818      0.078     -3.594      0.000      -0.435      -0.128
C(department)[T.DEPARTMENT OF FINANCE]:C(job_level)[T.Other]
-0.3579      0.067     -5.368      0.000      -0.489      -0.227
C(department)[T.DEPARTMENT OF FLEET AND FACILITY MANAGEMENT]:C(job_level)[T.Other]
-0.3479      0.086     -4.044      0.000      -0.517      -0.179
C(department)[T.DEPARTMENT OF HOUSING]:C(job_level)[T.Other]
-0.3042      0.197     -1.543      0.123      -0.691       0.082
C(department)[T.DEPARTMENT OF HUMAN RESOURCES]:C(job_level)[T.Other]
-0.2757      0.102     -2.691      0.007      -0.476      -0.075
C(department)[T.DEPARTMENT OF LAW]:C(job_level)[T.Other]
-0.2891      0.096     -3.012      0.003      -0.477      -0.101
C(department)[T.DEPARTMENT OF PLANNING AND DEVELOPMENT]:C(job_level)[T.Other]
-0.1403      0.127     -1.105      0.269      -0.389       0.109
C(department)[T.DEPARTMENT OF PROCUREMENT SERVICES]:C(job_level)[T.Other]
-0.0046      0.108     -0.043      0.966      -0.216       0.207
C(department)[T.DEPARTMENT OF STREETS AND SANITATION]:C(job_level)[T.Other]
-0.2380      0.071     -3.373      0.001      -0.376      -0.100
C(department)[T.DEPARTMENT OF TECHNOLOGY AND INNOVATION]:C(job_level)[T.Other]
0.0850      0.074      1.144      0.253      -0.061       0.231
C(department)[T.DEPARTMENT OF WATER MANAGEMENT]:C(job_level)[T.Other]
-0.2145      0.074     -2.898      0.004      -0.360      -0.069
C(department)[T.LICENSE APPEAL COMMISSION]:C(job_level)[T.Other]
0.1656      0.095      1.735      0.083      -0.021       0.353
C(department)[T.MAYORS OFFICE FOR PEOPLE WITH DISABILITIES]:C(job_level)[T.Other]
-0.2664      0.077     -3.473      0.001      -0.417      -0.116
C(department)[T.OFFICE OF BUDGET & MANAGEMENT]:C(job_level)[T.Other]
0.2416      0.078      3.091      0.002       0.088       0.395
C(department)[T.OFFICE OF CITY CLERK]:C(job_level)[T.Other]
-0.6282      0.086     -7.339      0.000      -0.796      -0.460
C(department)[T.OFFICE OF EMERGENCY MANAGEMENT AND COMMUNICATIONS]:C(job_level)[T.Other]
-0.2017      0.078     -2.595      0.009      -0.354      -0.049
C(department)[T.OFFICE OF INSPECTOR GENERAL]:C(job_level)[T.Other]
-0.5526      0.111     -4.962      0.000      -0.771      -0.334
C(department)[T.OFFICE OF PUBLIC SAFETY ADMINISTRATION]:C(job_level)[T.Other]
-0.4266      0.099     -4.307      0.000      -0.621      -0.232
C(department)[T.OFFICE OF THE MAYOR]:C(job_level)[T.Other]
-0.1169      0.061     -1.922      0.055      -0.236       0.002
C(department)[T.BOARD OF ETHICS]:C(job_level)[T.Professional]
1.692e-17   4.61e-17      0.367      0.714   -7.34e-17    1.07e-16
C(department)[T.CHICAGO ANIMAL CARE AND CONTROL]:C(job_level)[T.Professional]
1.693e-16   6.31e-17      2.684      0.007    4.57e-17    2.93e-16
C(department)[T.CHICAGO COMMISSION ON HUMAN RELATIONS]:C(job_level)[T.Professional]
-9.623e-17   5.72e-17     -1.682      0.093   -2.08e-16    1.59e-17
C(department)[T.CHICAGO DEPARTMENT OF AVIATION]:C(job_level)[T.Professional]
-0.1570      0.191     -0.823      0.411      -0.531       0.217
C(department)[T.CHICAGO DEPARTMENT OF PUBLIC HEALTH]:C(job_level)[T.Professional]
-0.1773      0.193     -0.916      0.359      -0.556       0.202
C(department)[T.CHICAGO DEPARTMENT OF TRANSPORTATION]:C(job_level)[T.Professional]
-0.2520      0.193     -1.308      0.191      -0.630       0.126
C(department)[T.CHICAGO FIRE DEPARTMENT]:C(job_level)[T.Professional]
-0.2904      0.215     -1.348      0.178      -0.713       0.132
C(department)[T.CHICAGO POLICE BOARD]:C(job_level)[T.Professional]
6.894e-17   4.28e-17      1.612      0.107   -1.49e-17    1.53e-16
C(department)[T.CHICAGO POLICE DEPARTMENT]:C(job_level)[T.Professional]
-0.4765      0.188     -2.532      0.011      -0.845      -0.108
C(department)[T.CHICAGO PUBLIC LIBRARY]:C(job_level)[T.Professional]
-0.0293      0.189     -0.156      0.876      -0.399       0.340
C(department)[T.CITY COUNCIL]:C(job_level)[T.Professional]
-0.5514      0.229     -2.408      0.016      -1.000      -0.102
C(department)[T.CITY TREASURER'S OFFICE]:C(job_level)[T.Professional]
-0.0569      0.181     -0.314      0.753      -0.411       0.298
C(department)[T.CIVILIAN OFFICE OF POLICE ACCOUNTABILITY]:C(job_level)[T.Professional]
-0.3120      0.229     -1.363      0.173      -0.761       0.137
C(department)[T.COMMUNITY COMMISSION FOR PUBLIC SAFETY AND
ACCOUNTABILITY]:C(job_level)[T.Professional]           -0.9164      0.321     -2.859      0.004      -
1.545      -0.288
C(department)[T.DEPARTMENT OF ADMINISTRATIVE HEARING]:C(job_level)[T.Professional]
-7.751e-17   3.41e-17     -2.274      0.023   -1.44e-16   -1.07e-17
C(department)[T.DEPARTMENT OF BUILDINGS]:C(job_level)[T.Professional]
-0.2273      0.200     -1.135      0.256      -0.620       0.165
C(department)[T.DEPARTMENT OF BUSINESS AFFAIRS AND CONSUMER PROTECTION]:C(job_level)[T.Professional]
-0.3337      0.229     -1.455      0.146      -0.783       0.116
C(department)[T.DEPARTMENT OF CULTURAL AFFAIRS AND SPECIAL EVENTS]:C(job_level)[T.Professional]
2.218e-17   2.26e-17      0.980      0.327   -2.22e-17    6.65e-17
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C(department)[T.DEPARTMENT OF ENVIRONMENT]:C(job_level)[T.Professional]
-0.1208 0.193 -0.626 0.531 -0.499 0.258
C(department)[T.DEPARTMENT OF FAMILY AND SUPPORT SERVICES]:C(job_level)[T.Professional]
-0.5248 0.194 -2.706 0.007 -0.905 -0.145
C(department)[T.DEPARTMENT OF FINANCE]:C(job_level)[T.Professional]
-0.1571 0.190 -0.826 0.409 -0.530 0.216
C(department)[T.DEPARTMENT OF FLEET AND FACILITY MANAGEMENT]:C(job_level)[T.Professional]
-0.3684 0.212 -1.738 0.082 -0.784 0.047
C(department)[T.DEPARTMENT OF HOUSING]:C(job_level)[T.Professional]
-0.2133 0.265 -0.804 0.421 -0.733 0.306
C(department)[T.DEPARTMENT OF HUMAN RESOURCES]:C(job_level)[T.Professional]
-0.6802 0.204 -3.328 0.001 -1.081 -0.280
C(department)[T.DEPARTMENT OF LAW]:C(job_level)[T.Professional]
-0.4342 0.237 -1.831 0.067 -0.899 0.031
C(department)[T.DEPARTMENT OF PLANNING AND DEVELOPMENT]:C(job_level)[T.Professional]
-0.1877 0.217 -0.866 0.387 -0.613 0.237
C(department)[T.DEPARTMENT OF PROCUREMENT SERVICES]:C(job_level)[T.Professional]
-0.0997 0.207 -0.483 0.629 -0.505 0.305
C(department)[T.DEPARTMENT OF STREETS AND SANITATION]:C(job_level)[T.Professional]
-0.2379 0.193 -1.234 0.217 -0.616 0.140
C(department)[T.DEPARTMENT OF TECHNOLOGY AND INNOVATION]:C(job_level)[T.Professional]
-0.0641 0.171 -0.375 0.707 -0.399 0.270
C(department)[T.DEPARTMENT OF WATER MANAGEMENT]:C(job_level)[T.Professional]
-0.1517 0.192 -0.790 0.430 -0.528 0.225
C(department)[T.LICENSE APPEAL COMMISSION]:C(job_level)[T.Professional]
-8.416e-17 2.81e-17 -2.999 0.003 -1.39e-16 -2.92e-17
C(department)[T.MAYORS OFFICE FOR PEOPLE WITH DISABILITIES]:C(job_level)[T.Professional]
-0.1257 0.181 -0.694 0.488 -0.481 0.229
C(department)[T.OFFICE OF BUDGET & MANAGEMENT]:C(job_level)[T.Professional]
-0.2903 0.167 -1.743 0.081 -0.617 0.036
C(department)[T.OFFICE OF CITY CLERK]:C(job_level)[T.Professional]
-0.6257 0.234 -2.678 0.007 -1.084 -0.168
C(department)[T.OFFICE OF EMERGENCY MANAGEMENT AND COMMUNICATIONS]:C(job_level)[T.Professional]
-0.4591 0.229 -2.006 0.045 -0.908 -0.010
C(department)[T.OFFICE OF INSPECTOR GENERAL]:C(job_level)[T.Professional]
-0.9089 0.207 -4.394 0.000 -1.314 -0.503
C(department)[T.OFFICE OF PUBLIC SAFETY ADMINISTRATION]:C(job_level)[T.Professional]
-0.6363 0.204 -3.116 0.002 -1.036 -0.236
C(department)[T.OFFICE OF THE MAYOR]:C(job_level)[T.Professional]
-0.2688 0.169 -1.595 0.111 -0.599 0.062
C(department)[T.BOARD OF ETHICS]:C(job_level)[T.Senior Management]
-0.2176 0.156 -1.397 0.162 -0.523 0.088
C(department)[T.CHICAGO ANIMAL CARE AND CONTROL]:C(job_level)[T.Senior Management]
-0.3447 0.145 -2.371 0.018 -0.630 -0.060
C(department)[T.CHICAGO COMMISSION ON HUMAN RELATIONS]:C(job_level)[T.Senior Management]
0 0 nan nan 0 0
C(department)[T.CHICAGO DEPARTMENT OF AVIATION]:C(job_level)[T.Senior Management]
-0.3376 0.079 -4.287 0.000 -0.492 -0.183
C(department)[T.CHICAGO DEPARTMENT OF PUBLIC HEALTH]:C(job_level)[T.Senior Management]
-0.4461 0.084 -5.338 0.000 -0.610 -0.282
C(department)[T.CHICAGO DEPARTMENT OF TRANSPORTATION]:C(job_level)[T.Senior Management]
-0.4626 0.095 -4.885 0.000 -0.648 -0.277
C(department)[T.CHICAGO FIRE DEPARTMENT]:C(job_level)[T.Senior Management]
-0.3281 0.128 -2.562 0.010 -0.579 -0.077
C(department)[T.CHICAGO POLICE BOARD]:C(job_level)[T.Senior Management]
0 0 nan nan 0 0
C(department)[T.CHICAGO POLICE DEPARTMENT]:C(job_level)[T.Senior Management]
-0.2223 0.074 -3.010 0.003 -0.367 -0.078
C(department)[T.CHICAGO PUBLIC LIBRARY]:C(job_level)[T.Senior Management]
-0.3541 0.109 -3.245 0.001 -0.568 -0.140
C(department)[T.CITY COUNCIL]:C(job_level)[T.Senior Management]
0 0 nan nan 0 0
C(department)[T.CITY TREASURER'S OFFICE]:C(job_level)[T.Senior Management]
-0.1503 0.089 -1.685 0.092 -0.325 0.025
C(department)[T.CIVILIAN OFFICE OF POLICE ACCOUNTABILITY]:C(job_level)[T.Senior Management]
-0.5995 0.236 -2.539 0.011 -1.062 -0.137
C(department)[T.COMMUNITY COMMISSION FOR PUBLIC SAFETY AND ACCOUNTABILITY]:C(job_level)[T.Senior
Management] -0.8276 0.208 -3.988 0.000 -1.234 -0.421
C(department)[T.DEPARTMENT OF ADMINISTRATIVE HEARING]:C(job_level)[T.Senior Management]
-0.2079 0.156 -1.337 0.181 -0.513 0.097
C(department)[T.DEPARTMENT OF BUILDINGS]:C(job_level)[T.Senior Management]
-0.5127 0.100 -5.119 0.000 -0.709 -0.316
C(department)[T.DEPARTMENT OF BUSINESS AFFAIRS AND CONSUMER PROTECTION]:C(job_level)[T.Senior
Management] -0.5559 0.166 -3.348 0.001 -0.881 -0.230
C(department)[T.DEPARTMENT OF CULTURAL AFFAIRS AND SPECIAL EVENTS]:C(job_level)[T.Senior Management]
-0.1296 0.095 -1.359 0.174 -0.317 0.057

C(department)[T.DEPARTMENT OF ENVIRONMENT]:C(job_level)[T.Senior Management]
-0.2792 0.104 -2.693 0.007 -0.482 -0.076
C(department)[T.DEPARTMENT OF FAMILY AND SUPPORT SERVICES]:C(job_level)[T.Senior Management]
-0.4688 0.088 -5.335 0.000 -0.641 -0.297
C(department)[T.DEPARTMENT OF FINANCE]:C(job_level)[T.Senior Management]
-0.0099 0.078 -0.128 0.898 -0.162 0.143
C(department)[T.DEPARTMENT OF FLEET AND FACILITY MANAGEMENT]:C(job_level)[T.Senior Management]
-0.4632 0.096 -4.835 0.000 -0.651 -0.275
C(department)[T.DEPARTMENT OF HOUSING]:C(job_level)[T.Senior Management]
-0.4118 0.208 -1.984 0.047 -0.819 -0.005
C(department)[T.DEPARTMENT OF HUMAN RESOURCES]:C(job_level)[T.Senior Management]
-0.8541 0.120 -7.114 0.000 -1.089 -0.619
C(department)[T.DEPARTMENT OF LAW]:C(job_level)[T.Senior Management]
-0.2212 0.101 -2.189 0.029 -0.419 -0.023
C(department)[T.DEPARTMENT OF PLANNING AND DEVELOPMENT]:C(job_level)[T.Senior Management]
-0.2308 0.152 -1.520 0.129 -0.528 0.067
C(department)[T.DEPARTMENT OF PROCUREMENT SERVICES]:C(job_level)[T.Senior Management]
0.0331 0.123 0.269 0.788 -0.208 0.274
C(department)[T.DEPARTMENT OF STREETS AND SANITATION]:C(job_level)[T.Senior Management]
-0.3877 0.120 -3.240 0.001 -0.622 -0.153
C(department)[T.DEPARTMENT OF TECHNOLOGY AND INNOVATION]:C(job_level)[T.Senior Management]
-0.2919 0.075 -3.896 0.000 -0.439 -0.145
C(department)[T.DEPARTMENT OF WATER MANAGEMENT]:C(job_level)[T.Senior Management]
-0.2458 0.103 -2.388 0.017 -0.448 -0.044
C(department)[T.LICENSE APPEAL COMMISSION]:C(job_level)[T.Senior Management]
0 0 nan nan 0 0
C(department)[T.MAYORS OFFICE FOR PEOPLE WITH DISABILITIES]:C(job_level)[T.Senior Management]
0 0 nan nan 0 0
C(department)[T.OFFICE OF BUDGET & MANAGEMENT]:C(job_level)[T.Senior Management]
0.0516 0.080 0.647 0.517 -0.105 0.208
C(department)[T.OFFICE OF CITY CLERK]:C(job_level)[T.Senior Management]
-0.4308 0.105 -4.108 0.000 -0.636 -0.225
C(department)[T.OFFICE OF EMERGENCY MANAGEMENT AND COMMUNICATIONS]:C(job_level)[T.Senior Management]
-0.0827 0.085 -0.968 0.333 -0.250 0.085
C(department)[T.OFFICE OF INSPECTOR GENERAL]:C(job_level)[T.Senior Management]
-0.7064 0.125 -5.666 0.000 -0.951 -0.462
C(department)[T.OFFICE OF PUBLIC SAFETY ADMINISTRATION]:C(job_level)[T.Senior Management]
-0.3889 0.108 -3.616 0.000 -0.600 -0.178
C(department)[T.OFFICE OF THE MAYOR]:C(job_level)[T.Senior Management]
0.0432 0.078 0.551 0.581 -0.110 0.197
C(department)[T.BOARD OF ETHICS]:C(job_level)[T.Senior Professional]
0 0 nan nan 0 0
C(department)[T.CHICAGO ANIMAL CARE AND CONTROL]:C(job_level)[T.Senior Professional]
0 0 nan nan 0 0
C(department)[T.CHICAGO COMMISSION ON HUMAN RELATIONS]:C(job_level)[T.Senior Professional]
0.1912 0.153 1.248 0.212 -0.109 0.491
C(department)[T.CHICAGO DEPARTMENT OF AVIATION]:C(job_level)[T.Senior Professional]
0.0652 0.066 0.981 0.327 -0.065 0.195
C(department)[T.CHICAGO DEPARTMENT OF PUBLIC HEALTH]:C(job_level)[T.Senior Professional]
-0.0769 0.068 -1.140 0.254 -0.209 0.055
C(department)[T.CHICAGO DEPARTMENT OF TRANSPORTATION]:C(job_level)[T.Senior Professional]
0.0207 0.080 0.258 0.796 -0.137 0.178
C(department)[T.CHICAGO FIRE DEPARTMENT]:C(job_level)[T.Senior Professional]
-0.1174 0.127 -0.922 0.356 -0.367 0.132
C(department)[T.CHICAGO POLICE BOARD]:C(job_level)[T.Senior Professional]
0 0 nan nan 0 0
C(department)[T.CHICAGO POLICE DEPARTMENT]:C(job_level)[T.Senior Professional]
-0.2163 0.055 -3.958 0.000 -0.323 -0.109
C(department)[T.CHICAGO PUBLIC LIBRARY]:C(job_level)[T.Senior Professional]
-0.0204 0.056 -0.364 0.716 -0.130 0.089
C(department)[T.CITY COUNCIL]:C(job_level)[T.Senior Professional]
0 0 nan nan 0 0
C(department)[T.CITY TREASURER'S OFFICE]:C(job_level)[T.Senior Professional]
0.3799 0.121 3.142 0.002 0.143 0.617
C(department)[T.CIVILIAN OFFICE OF POLICE ACCOUNTABILITY]:C(job_level)[T.Senior Professional]
-0.0263 0.230 -0.114 0.909 -0.478 0.425
C(department)[T.COMMUNITY COMMISSION FOR PUBLIC SAFETY AND ACCOUNTABILITY]:C(job_level)[T.Senior
Professional] -0.6945 0.218 -3.189 0.001 -1.121 -0.268
C(department)[T.DEPARTMENT OF ADMINISTRATIVE HEARING]:C(job_level)[T.Senior Professional]
0.1409 0.106 1.323 0.186 -0.068 0.350
C(department)[T.DEPARTMENT OF BUILDINGS]:C(job_level)[T.Senior Professional]
-0.0821 0.116 -0.710 0.478 -0.309 0.145
C(department)[T.DEPARTMENT OF BUSINESS AFFAIRS AND CONSUMER PROTECTION]:C(job_level)[T.Senior
Professional] 0.0010 0.142 0.007 0.995 -0.277 0.279
C(department)[T.DEPARTMENT OF CULTURAL AFFAIRS AND SPECIAL EVENTS]:C(job_level)[T.Senior Professional]
-0.1028 0.163 -0.632 0.528 -0.422 0.216

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C(department)[T.DEPARTMENT OF ENVIRONMENT]:C(job_level)[T.Senior Professional]
-0.0874      0.164     -0.532      0.595      -0.409       0.234
C(department)[T.DEPARTMENT OF FAMILY AND SUPPORT SERVICES]:C(job_level)[T.Senior Professional]
0.1007      0.075      1.340      0.180      -0.047       0.248
C(department)[T.DEPARTMENT OF FINANCE]:C(job_level)[T.Senior Professional]
0.2765      0.057      4.860      0.000       0.165       0.388
C(department)[T.DEPARTMENT OF FLEET AND FACILITY MANAGEMENT]:C(job_level)[T.Senior Professional]
0.0712      0.094      0.761      0.447      -0.112       0.255
C(department)[T.DEPARTMENT OF HOUSING]:C(job_level)[T.Senior Professional]
-0.0014      0.194     -0.007      0.994      -0.382       0.379
C(department)[T.DEPARTMENT OF HUMAN RESOURCES]:C(job_level)[T.Senior Professional]
-0.3269      0.139     -2.358      0.018      -0.599      -0.055
C(department)[T.DEPARTMENT OF LAW]:C(job_level)[T.Senior Professional]
0.1723      0.107      1.615      0.106      -0.037       0.381
C(department)[T.DEPARTMENT OF PLANNING AND DEVELOPMENT]:C(job_level)[T.Senior Professional]
0.0883      0.125      0.707      0.479      -0.156       0.333
C(department)[T.DEPARTMENT OF PROCUREMENT SERVICES]:C(job_level)[T.Senior Professional]
-0.1199      0.143     -0.841      0.401      -0.400       0.160
C(department)[T.DEPARTMENT OF STREETS AND SANITATION]:C(job_level)[T.Senior Professional]
-0.2470      0.075     -3.309      0.001      -0.393      -0.101
C(department)[T.DEPARTMENT OF TECHNOLOGY AND INNOVATION]:C(job_level)[T.Senior Professional]
0.4803      0.055      8.805      0.000       0.373       0.587
C(department)[T.DEPARTMENT OF WATER MANAGEMENT]:C(job_level)[T.Senior Professional]
-0.0327      0.081     -0.406      0.684      -0.191       0.125
C(department)[T.LICENSE APPEAL COMMISSION]:C(job_level)[T.Senior Professional]
0          0        nan        nan          0           0
C(department)[T.MAYORS OFFICE FOR PEOPLE WITH DISABILITIES]:C(job_level)[T.Senior Professional]
0.4030      0.162      2.488      0.013       0.085       0.720
C(department)[T.OFFICE OF BUDGET & MANAGEMENT]:C(job_level)[T.Senior Professional]
0.0765      0.076      1.001      0.317      -0.073       0.226
C(department)[T.OFFICE OF CITY CLERK]:C(job_level)[T.Senior Professional]
-0.2435      0.103     -2.371      0.018      -0.445      -0.042
C(department)[T.OFFICE OF EMERGENCY MANAGEMENT AND COMMUNICATIONS]:C(job_level)[T.Senior Professional]
0.2704      0.066      4.074      0.000       0.140       0.400
C(department)[T.OFFICE OF INSPECTOR GENERAL]:C(job_level)[T.Senior Professional]
-0.3646      0.115     -3.161      0.002      -0.591      -0.139
C(department)[T.OFFICE OF PUBLIC SAFETY ADMINISTRATION]:C(job_level)[T.Senior Professional]
0.1703      0.101      1.685      0.092      -0.028       0.368
C(department)[T.OFFICE OF THE MAYOR]:C(job_level)[T.Senior Professional]
0.4309      0.079      5.461      0.000       0.276       0.586
==============================================================================
Omnibus:                     2441.324   Durbin-Watson:                   1.951
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             6459.605
Skew:                          -0.559   Prob(JB):                         0.00
Kurtosis:                       5.225   Cond. No.                     6.19e+16
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